



SECP2613
SYSTEM ANALYSIS AND DESIGN
SECTION 1

ASSIGNMENT 1

LECTURER:

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Question 1

a. As the systems analyst for ALLFOODS, I would assess the feasibility of the systems project Ahmad is proposing by considering several factors. First of all is the technical feasibility. Ahmad suggests implementing the current system to make it more functional but Abu argues that the current system has no major technical issues yet. Next is the operational feasibility. It discusses about the availability of the system to operate well once it has been installed and whether it fulfils the user requirements. In the beginning of this situation, Abu argues that people might dislike the change of the system and a new system would be very disruptive to their colleagues. Moreover, for the economic feasibility, the costs needed in developing, implementing, and maintaining the new system might lead to substantial losses or economic crisis to the company.

b. Based on Abu's conversations, the operational feasibility of the proposed project seems to be low. The existing system appears to be functioning well, and users are yet still satisfied with its performance and does not experiencing any significant issues. Implementing a new system without addressing significant operational problems or user dissatisfaction could disrupt existing operations, frustration among employees and negatively impact productivity. Therefore, from an operational standpoint, there may be other ways to update the company's image without necessitating a complete overhaul of the system.

c. Economic feasibility refers to whether the proposed project is financially viable and whether the resulted benefits outweigh the costs. Ahmad's indicates that implementing a new system would require a huge investment and what if the implementation requires a very long period, it might cause losses to the company. So, the economic feasibility would need to be carefully evaluated. Considering Abu's feedback that the current system is functioning adequately, the economic feasibility of the proposed project may be questionable if the benefits are not substantial enough to balance with the costs. The risk the company should bear shows that the economic feasibility of the proposed project is still considered as low. Both costs and benefits in implementing this system must be taken into consideration.

d. Technological feasibility refers to the need of improving the technical requirements by implementing the proposed project and determine whether making a new system is essential. We need to concern about the necessary technology to implement the proposed system and whether it can be integrated effectively with the existing systems and infrastructure. Ahmad confirms that the technology to update the system is required indicating that the project is technologically feasible, but Abu highlighted that the current system is functioning well. In my opinion, I think the technology is quite adequate in this stage so i could conclude that the technological feasibility is low too.

e. Based on the discussion between Ahmad and Abu, I would not recommend a full-blown systems study at this stage. The primary motivation for the proposed project seems to be improving ALLFOODS' image and being perceived as a technologically advanced company,

rather than addressing specific operational problems or functional deficiencies in the existing system. If the existing system still meets the operational needs and user requirements, alternative solutions such as enhancing the company's marketing could be more cost-effective ways to improve ALLFOODS' image without disrupting critical business processes.

Question 2

No.	Activity	Predecessor	Duration
1.0	Research Phase	-	4 weeks
1.1	Conduct market research on the existing student registration system	-	2 weeks
1.2	Analyze the results	1.1	1 week
1.3	Examine the platforms and technologies available about registration system	1.2	1 week
2.0	Planning Phase	Research phase	4 weeks
2.1	Determine the essential features and functions for registration system	-	2 week
2.2	Define goals, deliverables and scopes	2.1	1 week
2.3	Draft document outlining system requirements	-	1 week
2.4	Set financial limits and resource distribution	2.2, 2.3	1 week
3.0	Design Phase	Planning phase	10 weeks
3.1	Identify development tools	-	1 week
3.2	Design database schema and system architecture	3.1	3 week
3.3	Create user interface (UI) and user experience (UX) suitable for students	3.2	6 weeks
3.4	Create a prototype and solicit user input	-	6 weeks
4.0	Implement Phase	Design phase	3 weeks
4.1	Analyze requirements to identify smaller coding tasks	-	3 weeks
5.0	Test Phase	Implement phase	2 weeks
5.1	Combine automation and manual testing to check for bugs	-	2 weeks
6.0	Deploy Phase	Test phase	1 week
6.1	Move latest build copy to the production environment such as packaging, environment, configuration and installation	-	1 week
7.0	Maintain Phase	Deploy phase	Ongoing
7.1	Fixes bugs and resolves customer issues	-	1 week
7.2	Manage software changes	-	Ongoing
7.3	Monitor overall system performance, security and user experience	-	Ongoing

QUESTION 3

Key Difference Managing Agile Projects vs. Traditional Projects by Manager

As the ongoing development of project management, remote teams and agile methodologies has become increasingly common. This report aims to highlight the key differences between agile project and traditional project that managers may encounter.

Communication Challenges

In traditional projects, team members often co-located that allowing them can have face-to-face communication and quick decision making. However, team members that involve in agile management projects are usually spread across different time zones and physical locations, leading to communication challenges. Managers need to leverage technology such as video conferencing, collaboration tools, and instant messaging platforms to foster the productive communication and maintain cohesiveness among teams.

Coordination Challenges

Traditional projects typically follow Waterfall Methodology which is a linear progression with tasks process sequentially. In contrast, agile projects embrace iterative development and require continuous collaboration and synchronization among dispersed teams. Managers need to employ frameworks such as Scrum and DevOps to manage workloads, assign tasks, and create continuous feedback loops to ensure alignment with project objectives.

Managing customers' expectations and communication

Both traditional and agile projects require effective communication to manage customers' expectations. However, in agile management projects, manager must proactively manage the expectations regarding the communication frequency and project deliverables. Proactive risk management, regular status updates, and clear communication can help to boost confidence and reduce the miscommunication.

Embracing Flexibility and adaptability

Unlike traditional projects with strict plans and fixed requirements, agile projects thrive on flexibility and adaptability. To enable teams to respond immediately to evolving customer needs and market analysis, manager must embrace change by fostering a culture of experimentation, continuous learning, and adaptation where teams are encouraged to iteratively enhance their output deliverables and incorporate feedback.

Conclusion:

To sum up, managing an agile management project brings unique challenges for managers compared to traditional projects. Communication, coordination, customers' expectations management and flexibility are essential elements for success in navigating the complexities of agile management project. Managers can foster innovation, collaborate effectively, and deliver value in a dynamic business environment by adopting proactive and adaptive approaches.

QUESTION 4

Q4 a) Cost-Benefit Analysis (CBA) $PV(n) = \text{Payment} \times (1/(1+C)^n)$				
Costs	Year 0	Year 1	Year 2	Year 3
<u>Development Costs</u>				
- Hardware	50000			
- Software	15000			
- Training	30000			
- Consulting	80000			
- Data Conversion	40000			
<u>Total</u>	215000			
<u>Production Costs</u>				
- Supplies		15000	15000	15000
- IS Supplies		75000	82500	90750
- Upgrades		10000	10000	10000
<u>Annual Prod Costs</u>		100000	107500	115750
(Present Value)		74074	58985	47046
<u>Accumulated Costs</u>		289074	348059	395105
Benefits	Year 0	Year 1	Year 2	Year 3
Improve Customer Service		150000	180000	216000
Increase Productivity		150000	187500	234375
<u>Total</u>		300000	367500	450375
Reduced inventory (Present Value) costs		222222	201646	183051
<u>Accumulated benefits</u>		222222	423868	606919
(Present Value)				
<u>Gain or Loss</u>		(66852)	75809	211814
b) Profitability Index $\frac{211814}{215000} = 0.99$ $\therefore$ Profitability index = 0.99 showing that is a not good investment because the profitability index is less than one				
$\hookrightarrow \frac{\text{Gain or Loss}}{\text{Development Cost}}$				

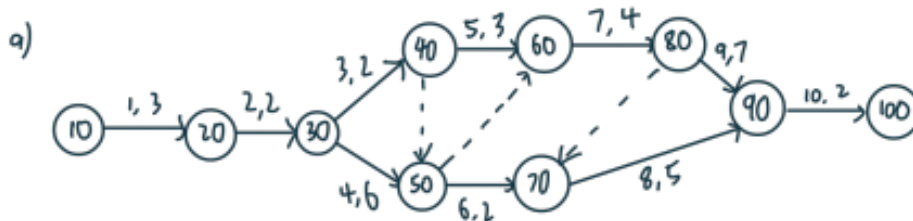
QUESTION 5

QUESTION 5 (10 MARKS)

A project has been defined to contain the following list of activities along with their required times of completion.

Activity No	Immediate Activity	Time (weeks)	Predecessors
1	Collect requirements	3	
2	Analyze processes	2	1
3	Analyze data	2	2
4	Design Process	6	2
5	Design data	3	3
6	Design Screens	2	3, 4
7	Design reports	4	4, 5
8	Program	5	6, 7
9	Test and document	7	7
10	Install	2	8, 9

- Draw a PERT Diagram for the activities. (3 Marks)
- Lists all the paths. (2 Marks)
- Calculate and identify the critical path. (2 Marks)
- Construct a Gantt chart for the project defined above. (3 Marks)



b) Path

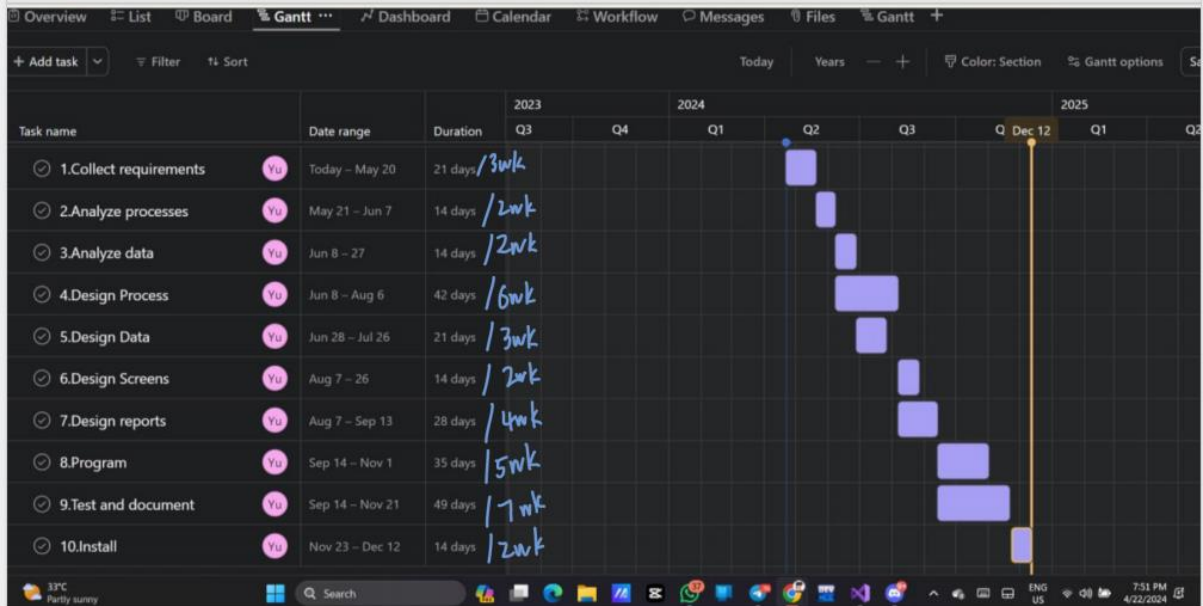
- Path 1 : 1-2-3-5-7-9-10
 Path 2 : 1-2-3-5-7-8-10
 Path 3 : 1-2-3-6-8-10
 Path 4 : 1-2-4-7-9-10
 Path 5 : 1-2-4-7-8-10
 Path 6 : 1-2-4-6-8-10

c) Length

- Path 1 : $3+2+2+3+4+7+2 = 23$ weeks
 Path 2 : $3+2+2+3+4+5+2 = 21$ weeks
 Path 3 : $3+2+2+2+5+2 = 16$ weeks
 Path 4 : $3+2+6+4+7+2 = 24$ weeks
 Path 5 : $3+2+6+4+5+2 = 22$ weeks
 Path 6 : $3+2+6+2+5+2 = 20$ weeks

∴ Path 4, 1-2-4-7-9-10 is the longest path, with a duration of 24 weeks.
 Hence, it is the longest path

d)



Gantt Chart Used in ASANA website